Keypad Based Product Counting and Automatic Batch Packing System with Station Sensing using LPC2129

ARM7 based embedded system | Two IR sensors, keypad, LCD and DC motor conveyor

1. Abstract

An LPC2129 (ARM7) microcontroller, two IR sensors and a keypad count products on a conveyor belt driven by a DC motor. The operator sets the batch quantity on the keypad and presses START. When the batch is complete, the motor stops and the LCD shows the final count. If a counted product never reaches the packing station, the system reports a belt fault.

The system uses two IR sensors (product counting and station check), one keypad for setting the batch quantity, and removes manual counting errors.

2. Objectives

1. **Automate counting:** count products on the belt using an external interrupt instead of counting by hand.
2. **Set batch size:** let the operator enter the batch quantity through the keypad.
3. **Detect belt faults:** identify a missing, fallen or blocked product at the packing station.
4. **Indicate status:** show the count on the LCD, with a buzzer and red / green LEDs.

3. Hardware Components

Component	Role
LPC2129	ARM7 main controller
4x4 Keypad	Batch quantity entry and START
IR Sensor 1	Counts products at the belt start
IR Sensor 2	Checks products at the packing station
16x2 LCD	Displays count and fault message
DC Motor	Drives the conveyor
Motor Driver	L293D or relay stage
Buzzer and LEDs	Red: running, Green: batch done

4. Block Diagram

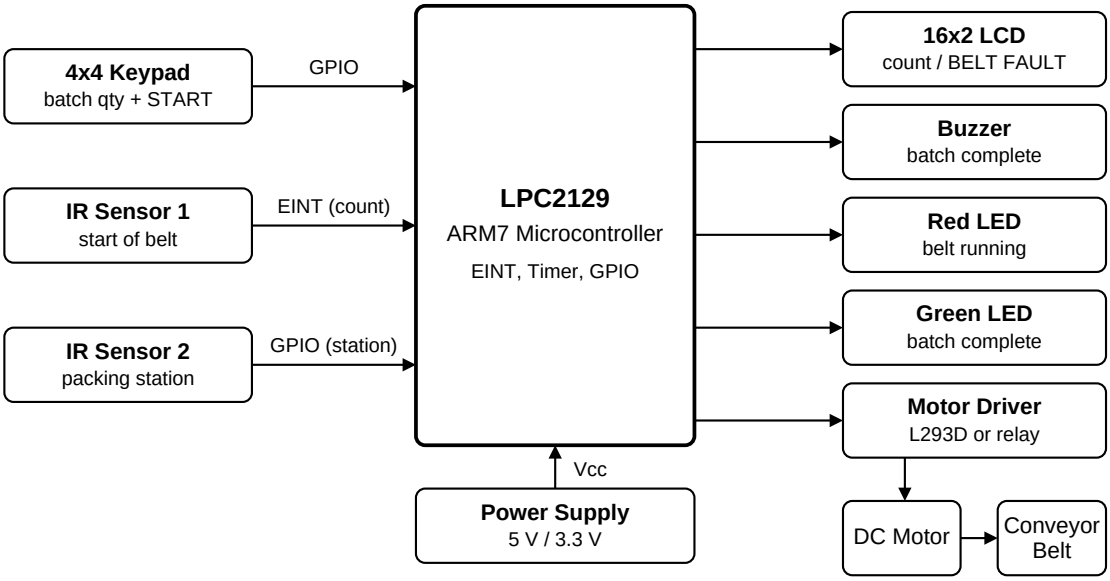

Figure 1. Block diagram of the system

5. Working Principle

1 The operator enters the batch quantity on the keypad and presses START.

2 The motor starts the conveyor and the red LED glows.

3 IR sensor 1 at the belt start raises an interrupt and the count increases.

4 IR sensor 2 at the packing station confirms that each product arrived.

5 If there is no detection within the set time, the motor stops and the LCD shows BELT FAULT.

6 When the count equals the batch quantity, the motor stops, the buzzer beeps, the green LED glows and the LCD shows the final count.

6. Flowchart

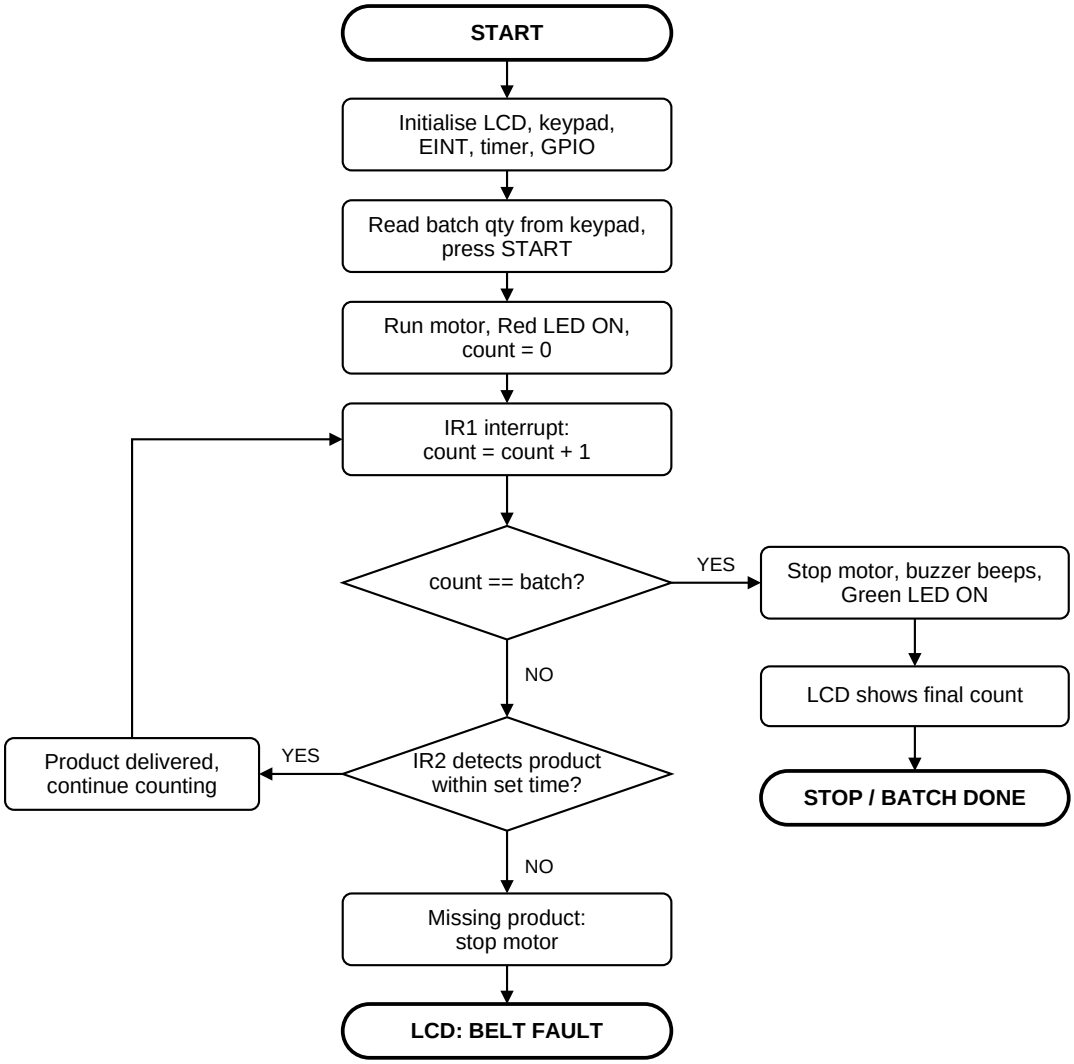

Figure 2. Flowchart of the working of the system

7. System Output Indications

Condition	Motor	LCD	LED	Buzzer
Belt running	ON	Live count	Red	Off
Batch complete	OFF	Final count	Green	Beeps
Missing product	OFF	BELT FAULT	Red off	Off

Fault logic: IR1 counted a product, but IR2 did not see it at the packing station within the set time.

8. Advantages and Applications

Advantages	Applications
Removes manual counting errors	Packaging lines
Detects fallen or blocked products	Bottling and carton filling
Low cost, simple ARM7 design	Production line batch counting
Clear LCD, LED and buzzer feedback	Warehouse conveyors

9. Conclusion

- The LPC2129 based system counts products accurately using an interrupt driven IR sensor.
- A second IR sensor at the packing station catches missing products and raises a BELT FAULT.
- It lowers manual counting errors and suits packaging and production lines.